WellSky® Blood Centers

Linux Systems Operations Guide

Software Version: 2026 R1

Release Date: November 2025

BCLXOP2026R1_01

How to Contact Us

WellSky Corporation
11300 Switzer Road
Overland Park, KS 66210

Customer Support: 866.714.9499

Fax: 630.684.0462

Web: http://www.wellsky.com

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Revision History

Software Release & Revision Date	Document Number	Content Affected	Description of Change
2026 R1November 2025	BCLXOP2026R1_01	All	Updated the document to reflect the current release version and release date.
2026 R1November 2025	BCLXOP2026R1_01	Section 3.4, Changing the Oracle Fusion Middleware Application Server Weblogic Password	Updated the command in step 15 to stop the application server. Updated the commands in step 20 to reference the BC_Domain. Updated the command in step 21 to start the application services.
2026 R1 November 2025	BCLXOP2026R1_01	Section 4.1, Oracle Shutdown	Replaced the screenshot in step 3, and updated the note to reference the PROD and Development databases. Updated the commands in steps 4, 5, and 8.
2026 R1 November 2025	BCLXOP2026R1_01	Section 4.2, Oracle Startup	Updated the commands in steps 5, 7, and 8.
2026 R1 November 2025	BCLXOP2026R1_01	Section 7.8, Purging Blood Centers Report Files	Updated the command in step 3.
2026 R1 November 2025	BCLXOP2026R1_01	Section 8.1.1, Cold Backup	Updated the database name in step 3, and updated the screenshot. Updated the directory in step 5.
2026 R1 November 2025	BCLXOP2026R1_01	Section 8.1.2, Hot Backup	Updated the commands in steps 6, 9, and 10.

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1.1 Overview

This guide provides information, procedures and recommendations for the operation of Linux database servers, client stations, and peripheral devices running WellSky Blood Centers applications.

Blood Centers customers should incorporate the operational requirements presented in this guide into their own operational procedures. Incorporating the stated requirements helps assure reliable system operation and compliance with regulatory requirements. Recommendations in this guide are based on experience and input received from customers and vendors. Customers should consider incorporating the recommendations in this guide into their operating procedures as appropriate for their individual operations. This guide does not serve as a substitute for the customer's own standard operating procedures.

1.2 Blood Centers Application Overview

Blood Centers is a Web-enabled application that supports three major subsystems: Donor, Lab, and Distribution.

Donor addresses mobile recruitment, drive scheduling, production control and transportation, donation registration, donor profile, donor communications, donor notification, drive analysis, collection staff scheduling, and HLA/RBC searches for matching donors, and special donations.

Each component contains appropriate interfaces with the other components as well as lab instrumentation in Lab.

Lab addresses all phases of lab functionality: receiving, component processing, testing, labeling, blood imports, special component processing, and centralized testing for multiple donor center testing.

Distribution addresses all phases of inventory and distribution: order entry and allocation, pricing, order processing, inventory management, packing and shipping, transfers, customer, and debit/credit.

1.3 System Configuration

Hardware configuration for Blood Centers addresses application-specific and customer-specific requirements. Blood Centers Professional Services provides customized configuration information for each customer as installations are planned and executed.

WellSky supports Blood Centers on Intel based systems running Oracle Enterprise Linux.

1.4 Data Base Management System

Blood Centers applications use the ORACLE Relational Database Management System (RDBMS). This high performance and fault-tolerant database management system offers the following benefits for online transaction processing applications:

- Easy access to all data
- Flexibility in data modeling
- Efficient data storage and redundancy

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- High-level data manipulation language (SQL)
- Portability to many other system platforms

The following are default names for the Blood Centers database instances:

Blood Centers Production	bcprd
Blood Centers Quality Assurance	bcqas
Blood Centers Training	bctrn

1.5 Disk Management

A typical Blood Centers system configuration uses disk mirroring or RAID for disk management. Mirroring is preferred to optimize performance and to control disk allocation for Oracle database files. Hardware RAID is preferred over software RAID in critical environments for efficiency and reliability. Auto-RAID environments are evaluated case-by-case. Configure auto-RAID with enough disk space to mirror the entire Blood Centers database for maximum operating efficiency.

1.6 Network Organization

1.6.1 Overview

A typical network hosting Blood Centers supports TCP/IP access over an Ethernet network running 100 MBPS and 1000 MBPS connections. Ethernet switches act as the network backbone. The switches provide dedicated high-speed connections between systems and user domains.

Selected systems act as Domain Name Servers (DNS) for the network. These systems perform name resolution, which allows other systems and clients to identify and access network entities by name without knowing or providing actual IP addresses.

Other devices on the network include Microsoft Windows servers, network printers, and PCs.

1.6.2 Wide Area Network (WAN)

Routers provide connections to a wide area network (WAN). A typical WAN is comprised of telecommunication circuits managed router-to-router to connect sites at different locations. T1 circuits typically provide both data transfer and long distance voice services. The WAN should be maintained as a private network with controlled connection to the Internet to protect critical systems and sensitive data.

Users may connect to the WAN over the Internet via VPN (Virtual Private Network) connections. VPN technology provides scrambled transmission of data over the Internet using high level encryption to ensure secure data transmission. Site managers must provide appropriate access controls when using public networks (VPN, dial-up, etc.) for access to the network.

1.6.3 Remote Sites

Typically remote sites connect to a primary Local Area Network (LAN) using routers and telecommunication circuits.

The remote site typically uses a local network hub connected to a router to establish a Local Area Network (LAN) for the remote site.

Devices connected to a remote site network include personal computers, print controllers, and printers.

Remote site PCs provide access to the Blood Centers systems and remote support. Remote users utilize various client software to connect to servers to access Blood Centers and other applications.

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2.1 Customer Facility Recommendations - Introduction

Each Customer should establish policies and procedures for a safe and secure operation. This "General Procedures" section includes general requirements and recommendations for Blood Centers customers.

2.2 System Manager Responsibilities

Customers should appoint a System Manager to be responsible for Blood Centers system management and computer operations. These tasks include:

2.2.1 Daily Tasks

- Run daily backups
- Run Mobile PC updates
- Check available disk space
- Check system logs
- Check error logs
- Verify that system components are online and operational

2.2.2 Weekly Tasks and Other Frequent Tasks

- Backup Quality Assurance (QA) databases
- Backup Training databases
- Backup Blood Centers application files
- Verify backup media
- Print donor ID cards

2.2.3 Monthly Tasks

Purge Blood Centers report files

2.2.4 Annual Tasks

Perform annual backups and database archiving



NOTE – These should be cold backups.

2.2.5 Database Maintenance

Schedule database maintenance quarterly, annually, or “as needed”, depending on the level of activity and available disk space on the system. Perform archiving and disk-maintenance activity according to a customer-defined maintenance schedule. The customer’s system manager should contact Customer Support to schedule these activities.

2.3 Standard Housekeeping Practices

1. Establish standard housekeeping practices to protect equipment, programs, and data.
2. Provide a safe and secure environment for Blood Centers systems including:
    - Maintain temperature and humidity in a safe operating range.
    - Keep the environment clean and dust free.
    - Protect media from damage due to scratching, static, etc.
    - Prohibit eating, drinking, or smoking near system components.
3. Properly train and supervise each Blood Centers system operator.
4. No operator should attempt to repair equipment without specific authorization of the supervisor or maintenance vendor.
5. No operator may alter programs or databases without specific authorization.
6. Do not use the Blood Centers system for other than assigned operations.

2.4 Security and Safety - Overview

Each Blood Centers customer computer facility should run as a closed shop. Blood Centers manages critical and confidential medical data. Each Blood Centers customer must take appropriate steps to protect confidential data and the integrity of the system to satisfy good manufacturing practices (cGMP), medical privacy requirements, and generally accepted business practices for systems and data security.

2.5 Security and Safety - Safeguards

The following safeguards should be in place for the security and safety of the computer center, the Blood Centers system, and its data:

- Restrict access to computer facilities to persons needed to operate, supervise or provide maintenance to the area and its equipment.
- Restrict critical functions to properly authorized managers and supervisors.
- Control access to systems and data. Require IDs and passwords. Terminate unused codes. Change passwords regularly. Require passwords for remote access.

- Provide reliable temperature and humidity controls. Backup environmental control units are desirable for critical operations.
- Install fire and smoke detection equipment. Fire suppression equipment should be installed for critical operations. Fire extinguishers should be available and easily accessible.
- Provide reliable conditioned power with surge suppression and UPS backup. A backup generator is desirable for critical operations.

2.6 General Maintenance

Unless specifically contracted otherwise, each Blood Centers customer maintains and supports Blood Centers system equipment with customer resources and vendor contracts for system hardware, operating system, and database management maintenance.

Maintenance includes both preventative and emergency maintenance for computer systems, hardware, and peripheral devices. Supplemental procedures for operation and maintenance of system devices are contained in vendor's manuals.

2.7 Preventive Maintenance

The only routine preventive maintenance needed for most Blood Centers system hardware is to clean tape drives using a cleaning cassette according to the manufacturer's recommendations. Consult your vendor or system documentation for any other special requirements for your system.

Cron jobs on Linux systems periodically trim log and trace files. Monitor the system routinely for growth of system logs and space availability.

2.8 Unlocking User Accounts



Important! The following procedure should only be done by a system administrator. Use this procedure only if the user cannot be unlocked through the Add System User (ADDUSER) screen.

If a user locks their account, you will need to unlock the account via SQL command. Follow the steps below to complete the unlock process.

1. Log onto the database server as **oracle**.
2. Right-click on the desktop and select **Open Terminal**.
3. Set the environment variables by entering the following command:
   `$ .oraenv`

   When prompted for the database, enter the name of the database where the user is locked (bcprd, bcqas, or bctrn).

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4. Enter the following command to log into the database:
   $ `sqlplus util/util`

   NOTE – If unlocking an account on the Mobile Laptop database, replace util with laptop.

5. A user's account is locked when the IS_PASSWORD_LOCKED field is set to Y. Verify the users account is locked by entering the following at the SQL prompt:
   SQL> `select USER_ID, IS_PASSWORD_LOCKED from SYSTEM_USERS where USER_ID='TRG001';`

   NOTE – Replace TRG001 with the user ID you want to unlock.

6. If the result of the query indicates a Y in the IS_PASSWORD_LOCKED field, continue with the remaining steps to unlock the account. If the IS_PASSWORD_LOCKED field is N, the user's account is not locked, so there is no need to continue with the remaining steps.

7. Unlock the account by entering the following at the SQL prompt:
   SQL> `update SYSTEM_USERS set IS_PASSWORD_LOCKED='N' where USER_ID='TRG001';`

   NOTE – Replace TRG001 with the user ID you want to unlock.

   SQL> `commit;`

8. Close the SQL session by entering the following at the SQL prompt:
   SQL> `exit;`

9. Close the terminal window.

10. Ask the user to log into the application to verify the account unlocked successfully.

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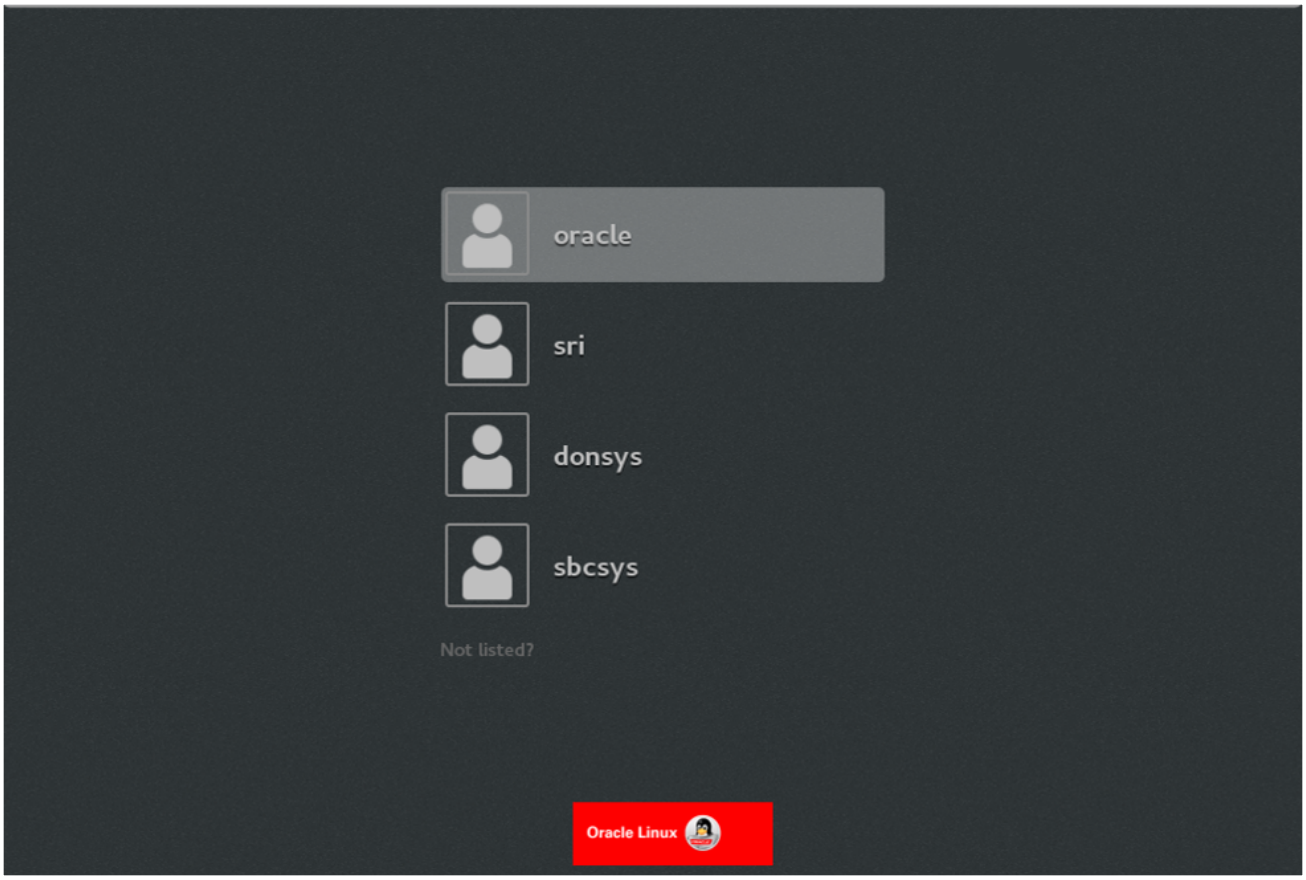
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3.1 Login and Logout of the Oracle Enterprise Linux System

3.1.1 Login Procedure

1. Logon to the system with your assigned ID.

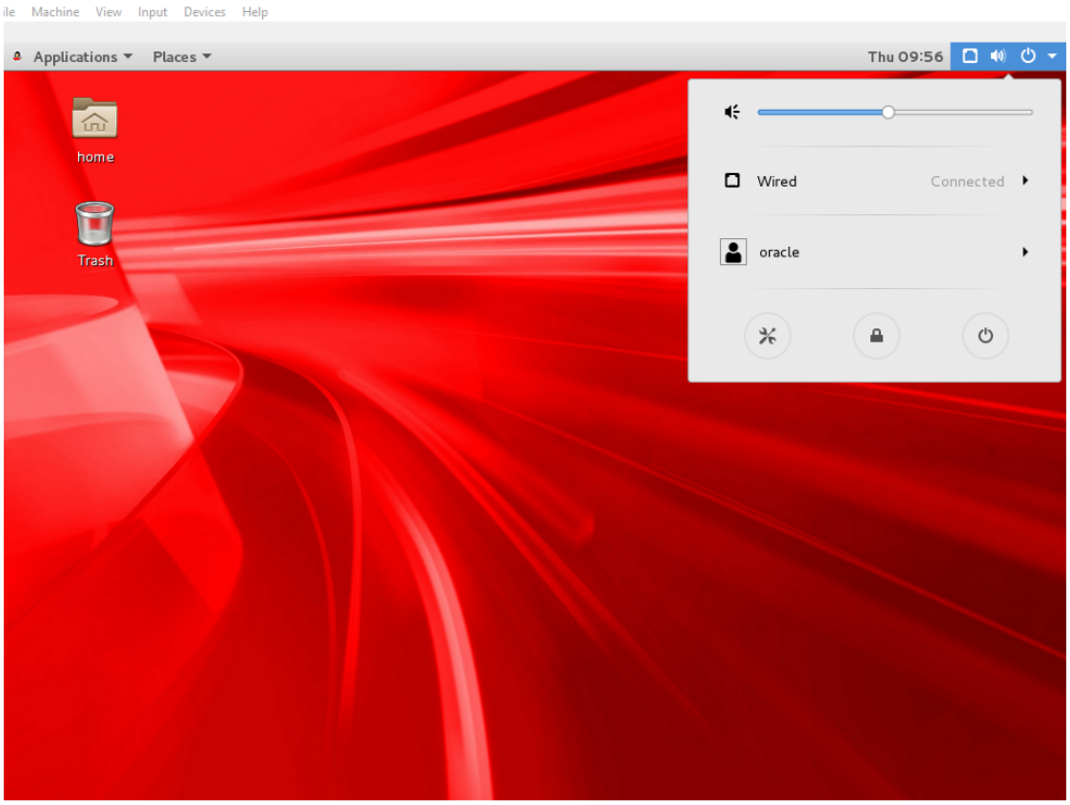


2. Enter your password.

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3.1.2 Logout Procedure

1. From the main menu at the top of the screen, select **System > Log Out <user currently logged in>**.



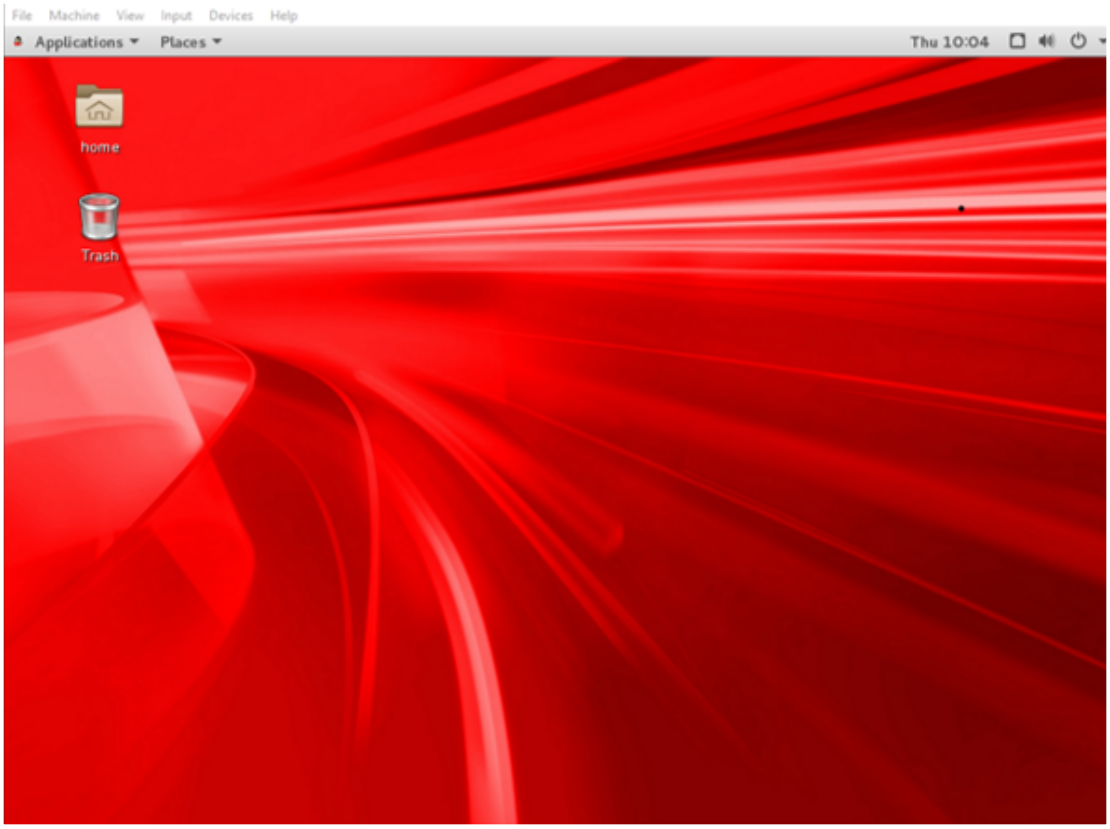
The Log out of this system now confirmation appears.



2. Click **Log Out** to logout immediately. If you do nothing, the log out will proceed after a 60 second countdown. To stop the log out process, click **Cancel**.

3.2 Linux Desktop

The figure below shows an example of an Oracle Enterprise Linux desktop. Your screen appearance may vary.



The menu at the top contains three items:

- Applications - used to browse and run installed applications
- Places - used to access documents, folders, and network places
- System- change desktop appearance and behavior, get help, or log out

The Blood Centers database server is dedicated to running the Oracle Database Management System and Blood Centers applications.

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3.3 Oracle Management Utilities

Oracle maintains three databases for Blood Centers: Production, Quality Assurance, and Training.

3.3.1 Oracle Enterprise Management Console

The Oracle Enterprise Management utility is used for managing the Oracle environment. To access the Enterprise Manager, follow these steps:

1. Open a Web browser and enter the following:
   `http://lifetrak.mediware.com:7001/em`

   NOTE – Replace lifetrak.mediware.com with the host name.domain name of your server.

2. You will be prompted for a user name and password. Enter as follows:

   Username = weblogic

   Password = WebLogicAdmin1

   NOTE – This is the default password. You should use the password you have set.

3.3.2 Oracle WebLogic Server Administration Console

To launch the Oracle WebLogic Server Administration Console, follow these steps:

1. Open a browser window and enter the following:
   `http://<server IP address>:7001/console`

   NOTE – Replace <server IP address> with the IP address of your Blood Centers server.

2. You will be prompted for a user name and password. Enter as follows:
   Username = weblogic

   Password = WebLogicAdmin1

   NOTE – This is the default password. You should use the password you have set.

3.4 Changing the Oracle Fusion Middleware Application Server Weblogic Password

Follow these steps to change the Weblogic password.

1. On the Blood Centers application server, log on as **oracle**.
2. Open a Web browser and enter the following:
   `http://<IP-ADDRESS>:7001/console`

   NOTE – Replace <IP-ADDRESS> with the IP Address of your server.

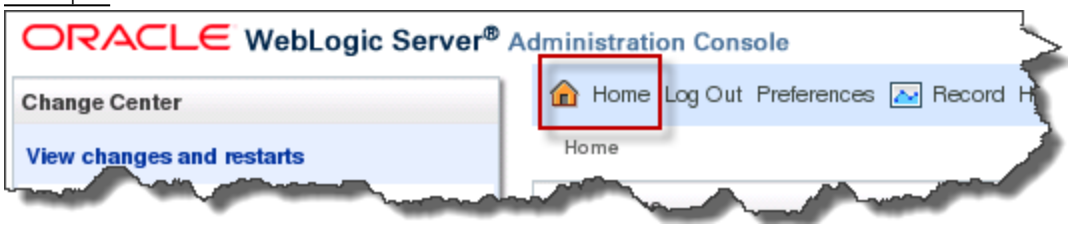
3. Log into the application by entering the following username and password.

   Username = weblogic

   Password = Your current password (the default password is WebLogicAdmin1)

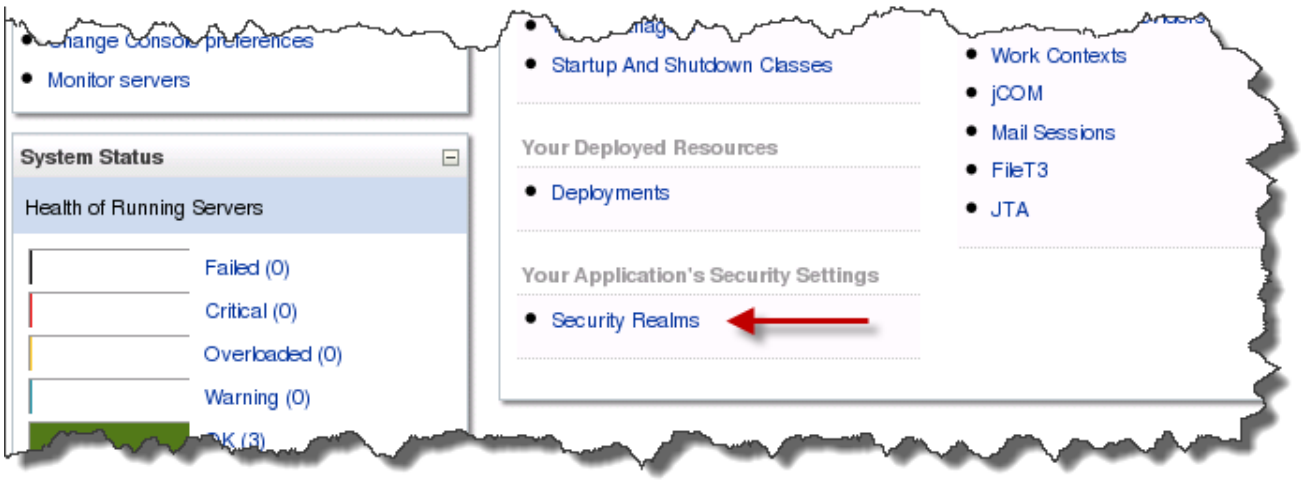
4. If the application does not open to the Home page, click **Home**.

   Example:



5. Scroll to the bottom of the page and click **Security Realms**.

   Example:



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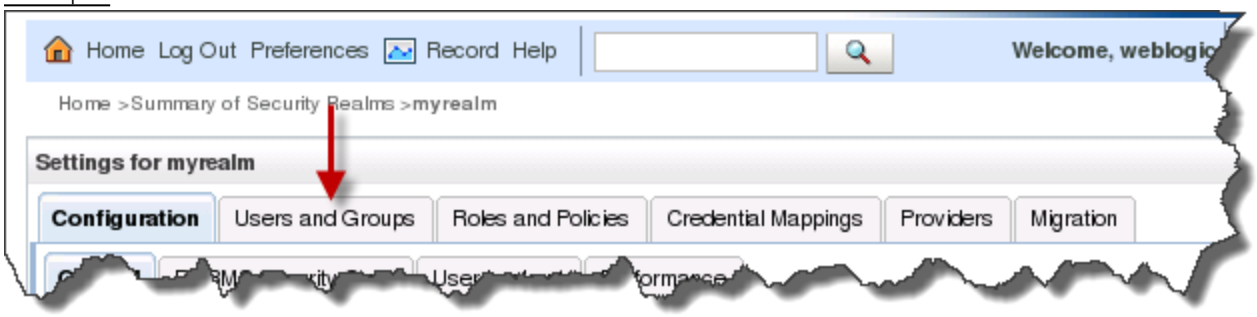
6. Click **myrealm**.

Example:



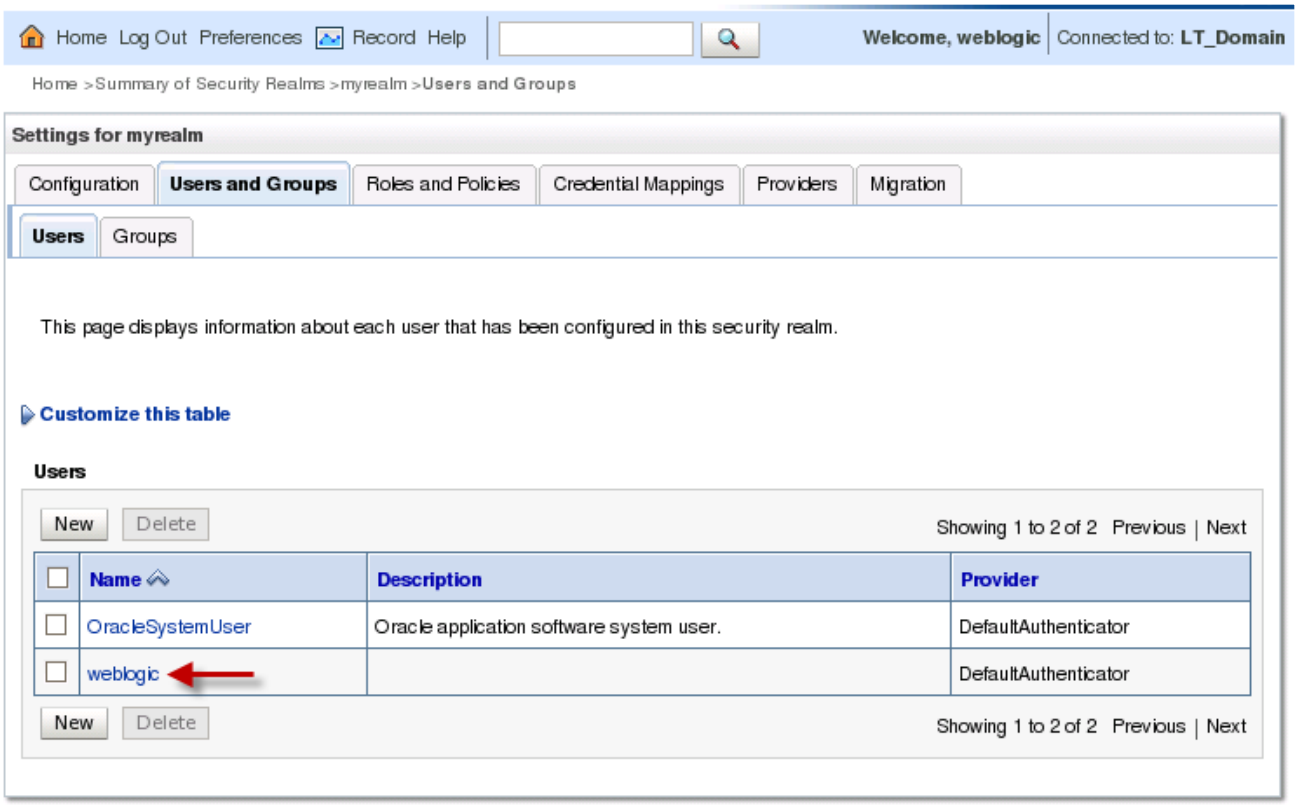
7. Click the **Users and Groups** tab.

Example:



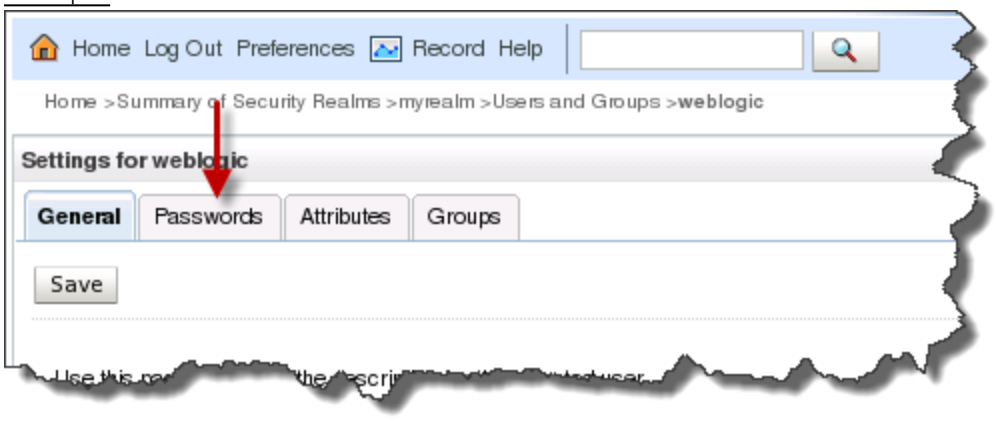
8. Click **weblogic**.

Example:



9. Click the **Passwords** tab.

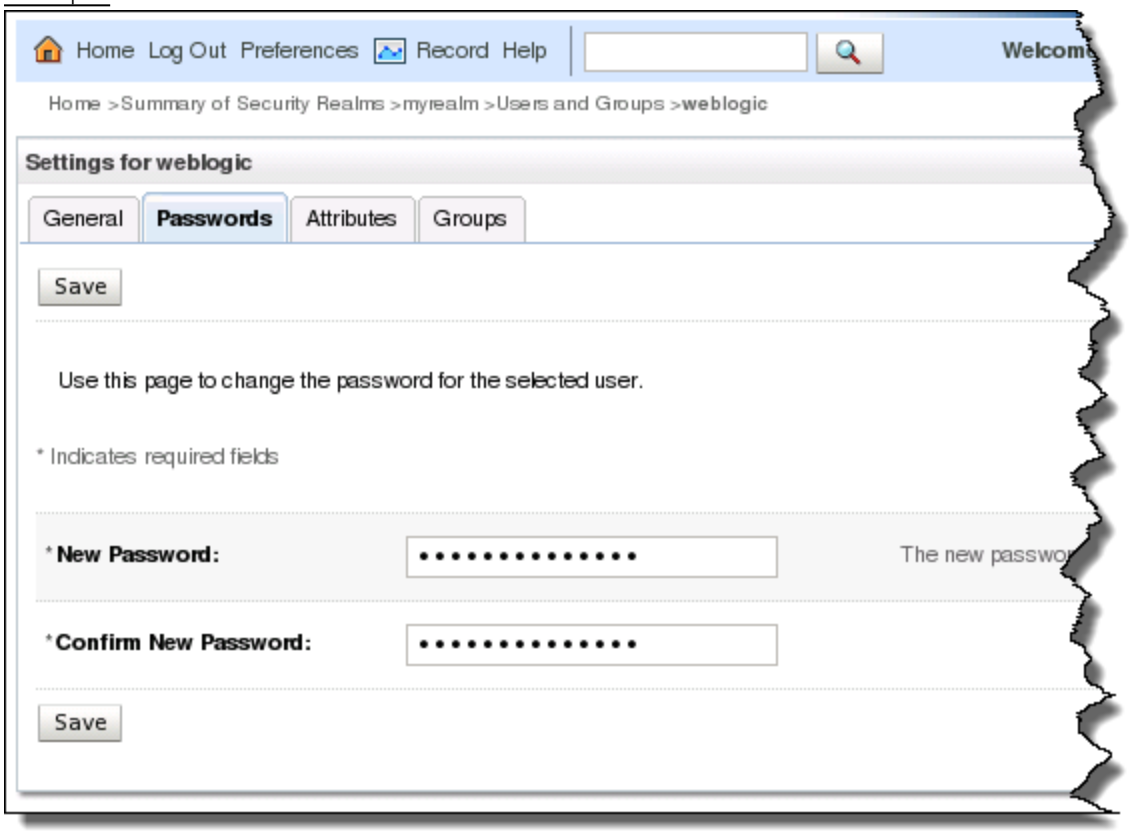
Example:



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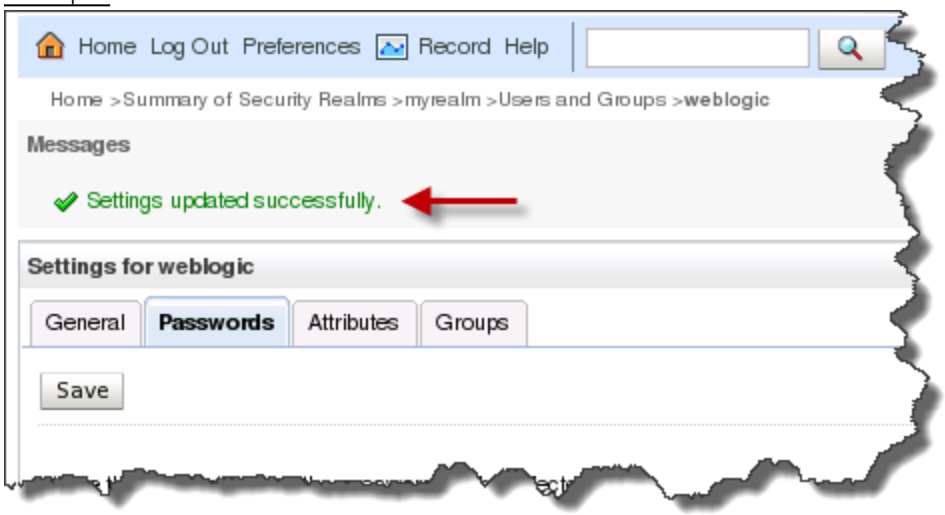
10. Enter the new password in the **Password** and **Confirm Password** boxes.

Example:



11. Click **Save**. A message appears indicating that the update was successful.

Example:



12. In the menu at the top next to Home, click **Logout**.

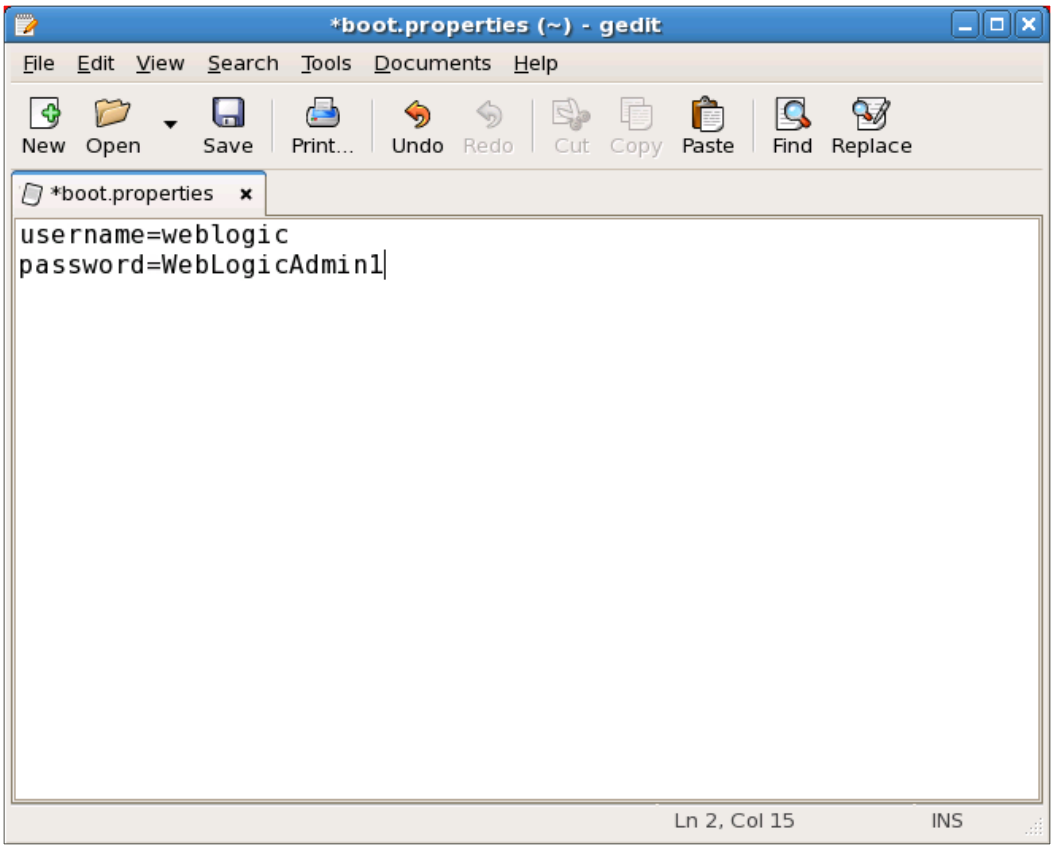
13. Close the browser.
14. Right-click on the desktop and select **Open Terminal**.
15. Stop the application server by entering the following command.
    $ `bcorappstop`

    Example:

    ```
    [oracle@ltappsvr01 ~]$  bcorappstop
    ```

16. Create a boot.properties file by entering the following command:
    $ `gedit boot.properties`
17. Enter `weblogic` on the first row.
18. Enter the new password you just set for weblogic on the second row.

    Example:



19. Save and close the file.

20. Copy the boot.properties file by entering the following commands:
    $ cp boot.properties /apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/servers/AdminServer/security
    $ cp boot.properties /apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/servers/WLS_FORMS/security
    $ cp boot.properties /apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/servers/WLS_REPORTS/security

    Example:

```
cp boot.properties /apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/servers/AdminServer/security
cp boot.properties /apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/servers/WLS_FORMS/security
cp boot.properties /apps/Middleware/Oracle_Home/user_projects/domains/BC_Domain/servers/WLS_REPORTS/security
```

    NOTE – Even though the commands shown in step 20 appear on multiple lines, there are no line breaks in the commands.

21. Start the application services by entering the following command:
    $ bcorappstart

    Example:

```
[oracle@ltappsvr01 ~]$ bcorappstart
```

If you encounter any weblogic password issues, please contact WellSky Support.

3.5 System and Oracle Database Utilities

3.5.1 Backups

Hot versus Cold – The Basics

For “hot” backups the databases are left running. Database recovery files are generated and archived to tape. For “cold” backups the databases are shutdown and all database files are archived to tape.

Backup Procedures

Production Databases

Backup production databases each business day to minimize data loss in the event of a system failure. Perform backups of Oracle Training and QA databases as needed. Monthly backup is recommended for training and QA databases.

System and Application Files

Backup system and application files at least weekly.

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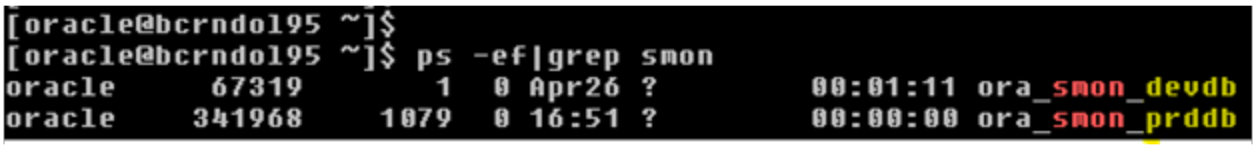
Chapter 4 - Oracle Startup and Shutdown

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4.1 Oracle Shutdown

1. Log on as **oracle**.
2. Right-click on the desktop and select **Open Terminal**.
3. Enter the following command:
   `$ ps -ef |grep smon`

   Example:



   NOTE – In the example above, the prddb database is for the PROD database, and devdb is for the Development database.

4. Set the Oracle application variables by entering the following command:
   `$ . bcfmwenv`
5. Shutdown all Oracle application services by entering the following command:
   `$ . bcorappstop`
6. Set the environment to the database you want to stop by entering the following command:
   `$ . oraenv`
7. When prompted for the database name, enter the name obtained in step 3.
8. Shutdown the specified Oracle database by entering the following command:
   `$ . bcoradbstop`
9. Verify the database shutdown by entering the following command:
   `$ ps -ef |grep smon`

   The database you just stopped should no longer be listed.
10. If you still have databases running, repeat steps 4-9. Once all databases have been stopped, continue to the next step.
11. Stop the Oracle database listener by entering the following command:
    `$ lsnrctl stop`
12. It is now safe to shutdown and reboot the server.

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4.2 Oracle Startup

1. Log in as **oracle**.
2. Set the database environment by entering the following command:
   `$ . oraenv`
3. When prompted for database name, enter the name of the Oracle database you wish to start (ex: tdonplp)
4. Start the Oracle database listener by entering the following command:
   `$ lsnrctl start`
5. Start the specified Oracle database by entering the following command:
   `$ . bcoradbstart`
6. Repeat steps 2-5 for each database you want to start.
7. Set the Oracle application variables by entering the following command:
   `$ . bcfmwenv`
8. Start all Oracle application services by entering the following command:
   `$ . bcorappstart`

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Chapter 5 - Managing Backup Media

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5.3 Off-Site Storage 28
5.4 Media Handling 28

5.1 Overview

Each site should establish and follow policies for backup media storage and handling. Media storage should include an off-site storage location for maximum protection with established procedures for rotating media to off-site storage.

5.2 Storage and Retention

Each site needs to establish and follow policies for storage and retention of backup media. Consideration should be given to the following items.

- Access to media storage should be controlled.
- Daily backups should be preserved for a minimum of one week to ensure recovery of timely data in case of a system failure.
- A media retention schedule should be defined that addresses daily, weekly, monthly, and annual backups to assure recovery of data when needed.
- Media retention and storage policies need to be clearly documented for staff, managers, regulators, and auditors to provide assurance that ongoing backup protection is in place.

5.3 Off-Site Storage

Each site should maintain a secure off-site storage facility with environmental controls to protect critical backups and media. Limit access to the storage facility. Critical operations should have round-the-clock access to off-site storage.

5.4 Media Handling

Good practices and procedures for handling media include:

- Avoid electrostatic discharge (ESD), radio (RF) transmission, and magnetic fields.
- Handle media with clean hands.
- Do not touch the surface of electromagnetic media.
- Keep media at or below room temperature.
- Use carrying containers when transporting media off-site.
- Mount all media carefully, precisely, and securely.
- Keep media in protective containers when not in use.
- Verify media is write-protected prior to reading.
- Verify label prior to mounting media for use.
- Maintain records of media used for backups.

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- Verify successful completion of backups.
- Segregate scratch media from media containing data.

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Chapter 6 - Remote Support

6.1 Customer Support 32

6.1 Customer Support

WellSky Customer Support provides remote support to Blood Centers customers.

Customer Support access can be provided through existing customer remote access mechanisms including:

- VPN (Virtual Private Network)
- Granting Internet firewall permissions
- RAS (Remote Access Server) connections
- SecureLink

A high-speed connection over the Internet is highly desirable because it eliminates long distance costs and expedites on-line support. This may be provided via VPN or by granting permissions for WellSky Customer Support to access the customer's Blood Centers systems through the customer's firewall.

Procedures for contacting WellSky Customer Support are provided in the Customer Service Guide.

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7

Chapter 7 - Blood Centers Printing

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7.1 Overview

This section provides guidance and instructions for Blood Centers printer operations. Refer to the Blood Centers installation guides for instructions on printer setup and printer specifications.

Most common Linux printer operations are handled from the Blood Centers operations menu. Printer management functions can also be handled from the Linux command line using standard Linux system commands.

7.2 Defining Printers in Blood Centers

Report printers must be defined in Blood Centers for Blood Centers automated printing. Only printers defined and named in the Blood Centers database can be used for automated printing.

For each Blood Centers user, set a default printer that is near the user's workstation. Reports may print to unexpected locations if no default printer is set for the user. The user's default printer may need to be redefined if the user moves.

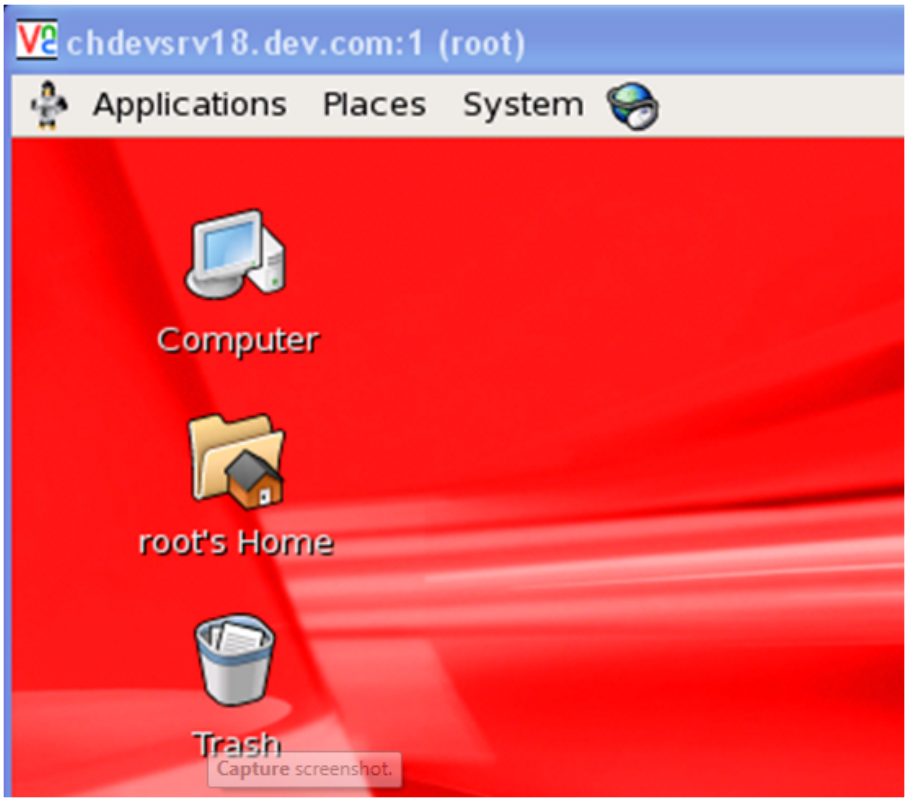
The Blood Centers System Administrator must also define default printers for warehouses defined in Blood Centers to route warehouse associated outputs to the correct printer.

Consult the Blood Centers User Guides for specific instructions to define printers, warehouses, and user defaults for Blood Centers applications.

7.3 Adding a Printer to the Blood Centers Linux Server

Use this procedure to add a network printer to the Blood Centers Linux server. The network printer must already have been added to your local network and properly configured.

1. Connect to the Linux Server using virtual network computing (VNC). Use your VNC password to log in. The following window opens:

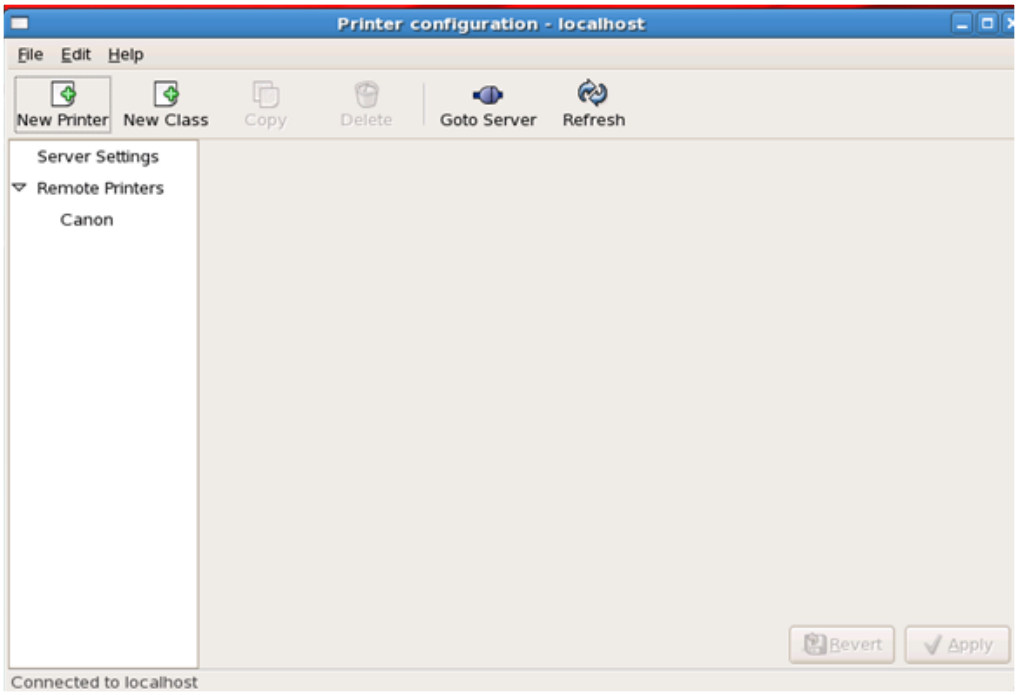


2. Click **System** > **Administration** > **Printing**, as shown:

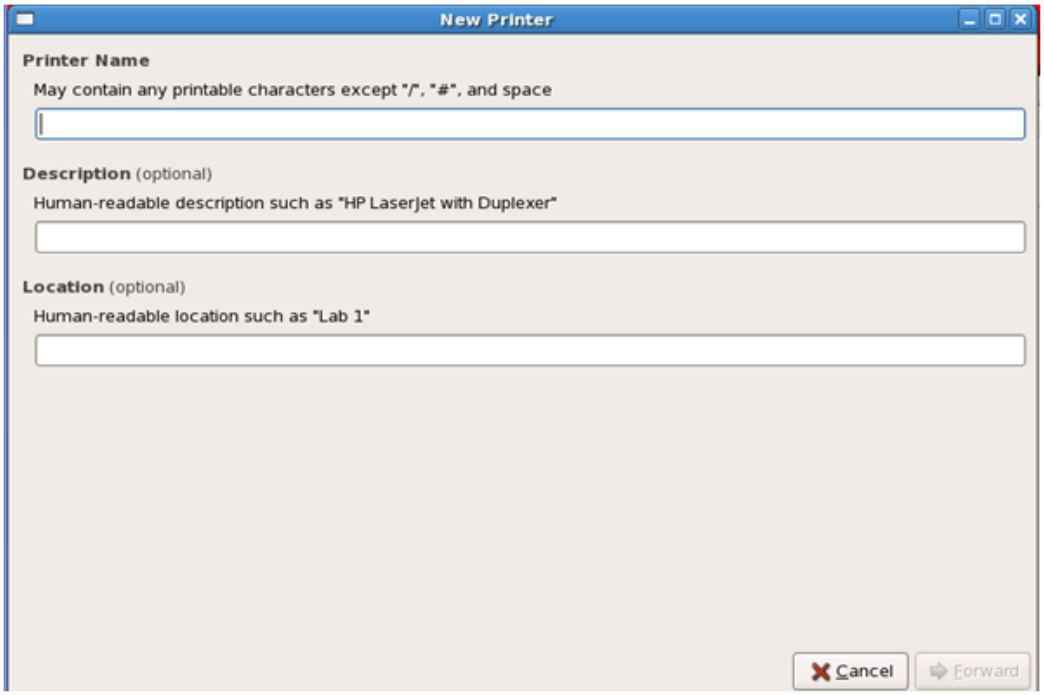




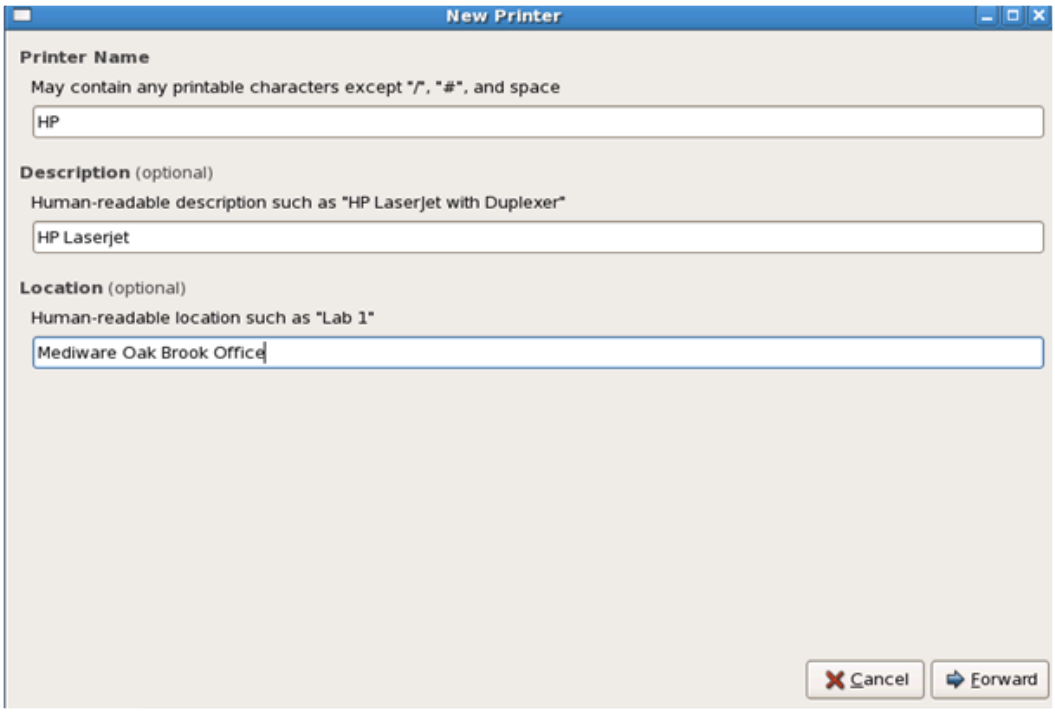
The following window opens:



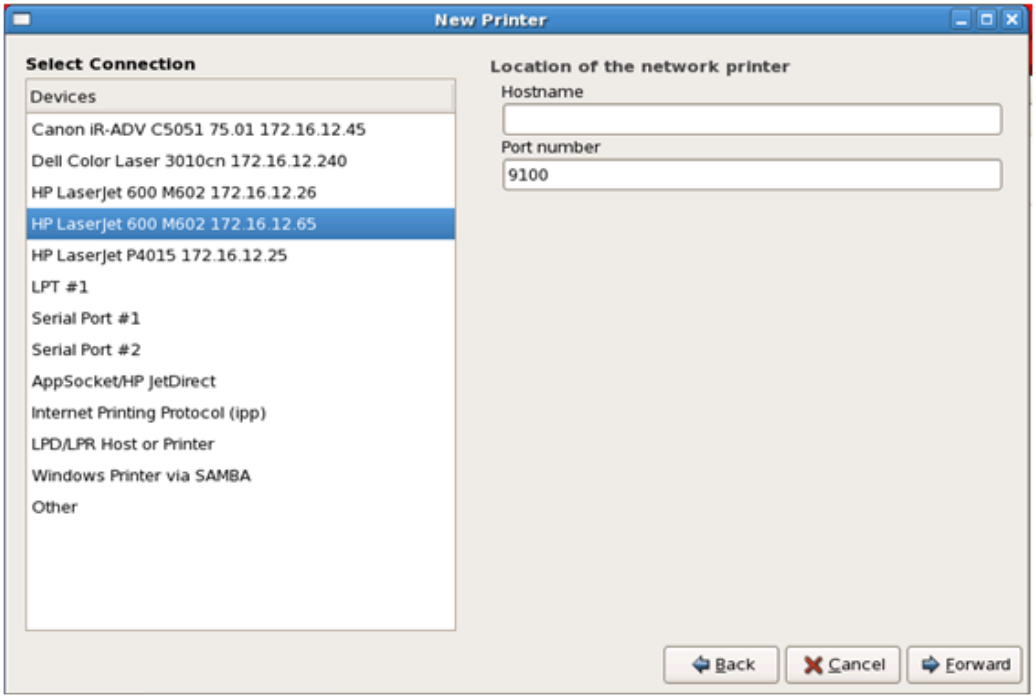
3. Click New Printer. The following window opens:



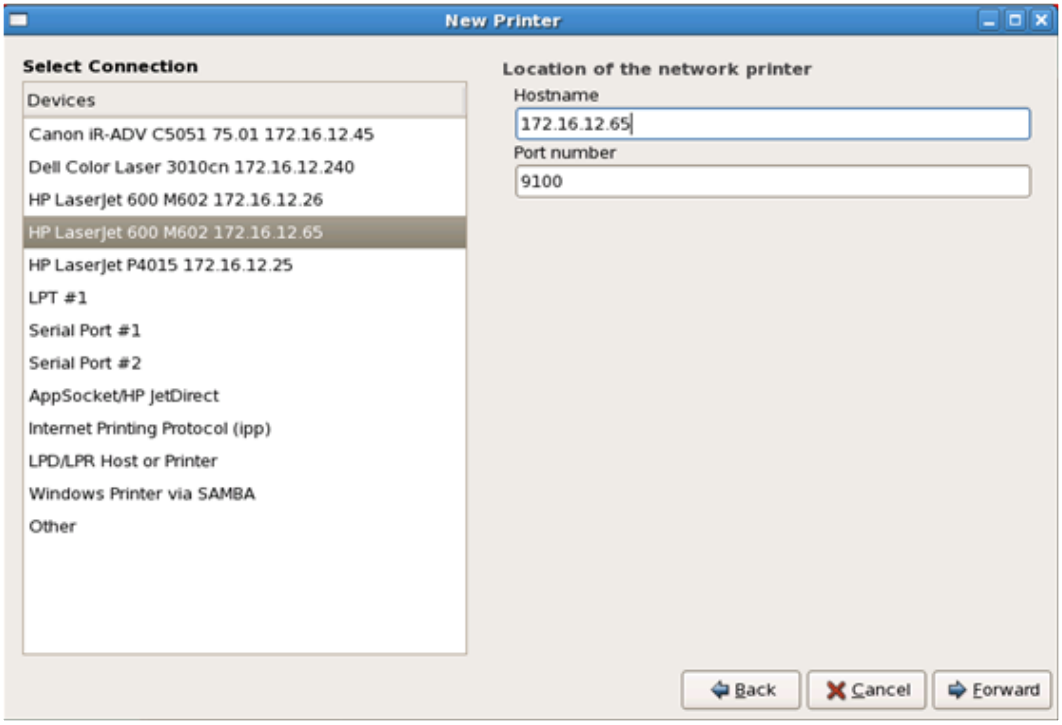
4. Enter the Printer Name, Printer Description (optional), and Printer Location (optional), as shown in the following example:



5. Click **Forward**. The following window opens:



6. Do the following. Refer to the example screen picture as needed.

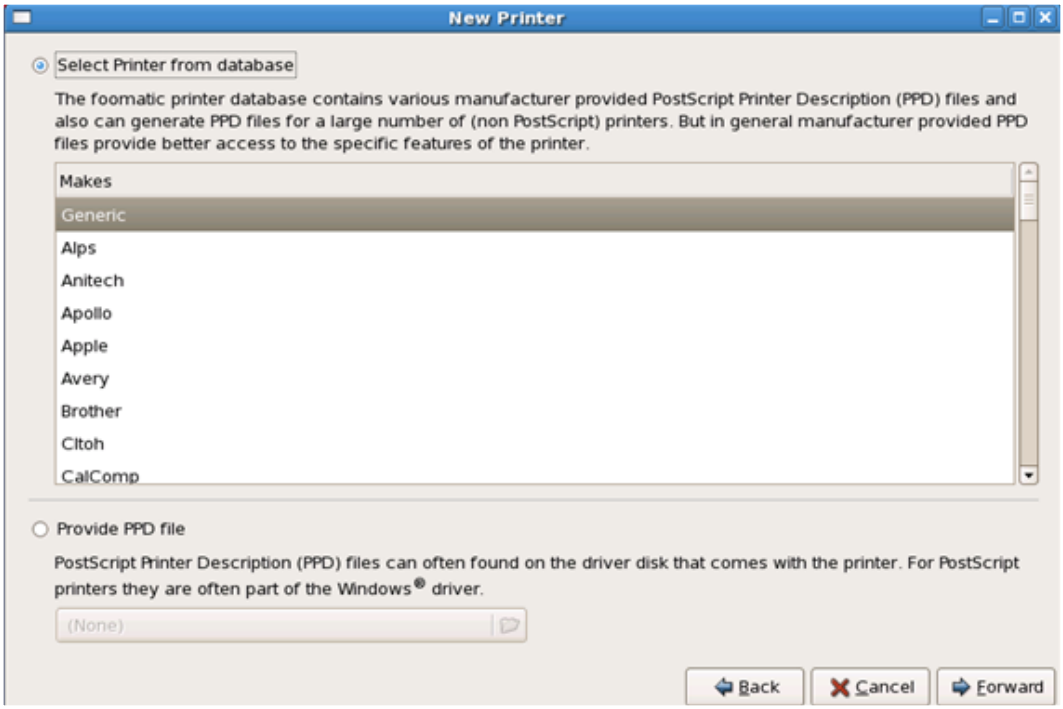


   a. Verify that the printer is listed in the Select Connection, Devices list. If not, consult your network or system administrator.

b. Enter the printer's IP address in the *Hostname* field.

c. Enter the Printer's port number in the *Port number* field.

7. Click **Forward**. The following window opens:

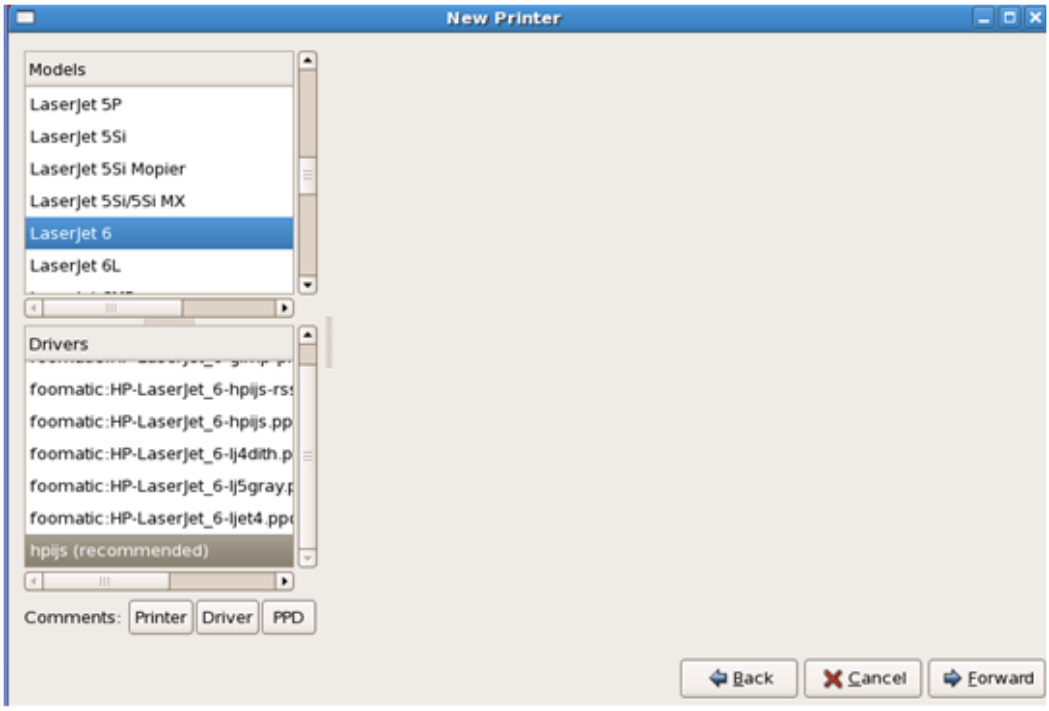


8. Do the following:

a. Select the *Select printer from a database* option.

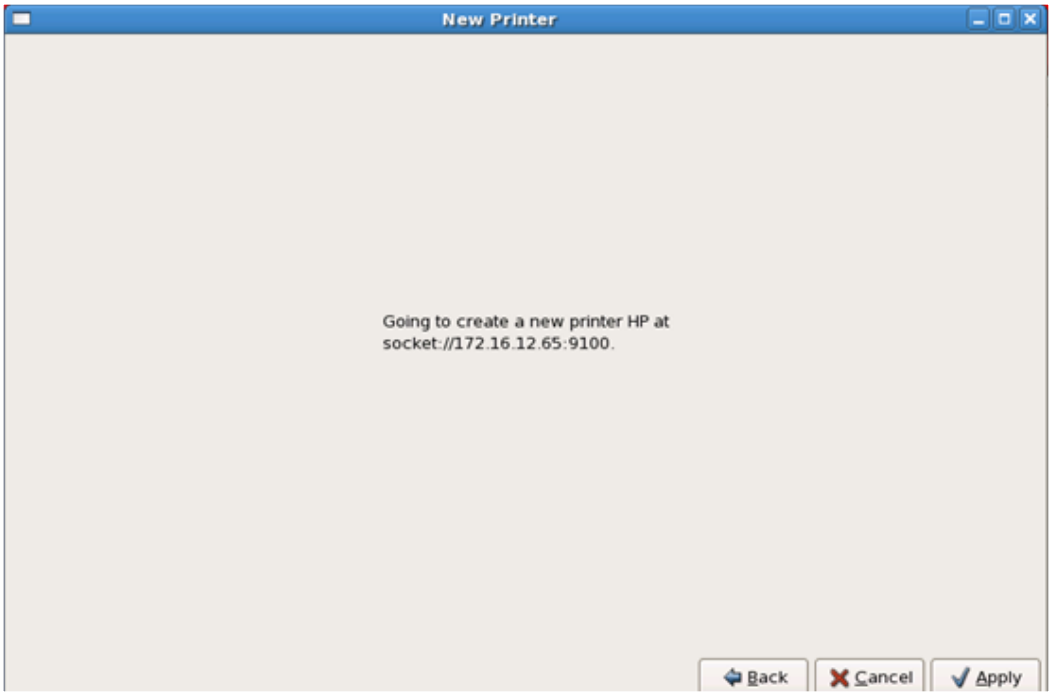
b. Locate and highlight your printer's make from the alphabetical *Makes* list.

9. Click **Forward**. The following window opens:

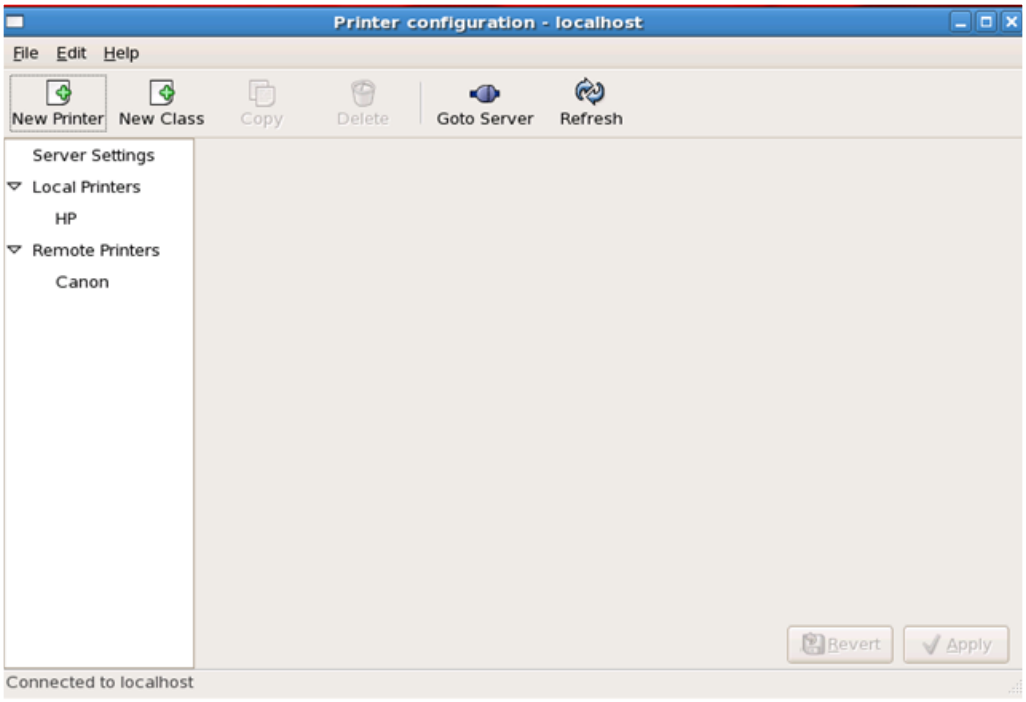


Locate and highlight the Printer's model number.

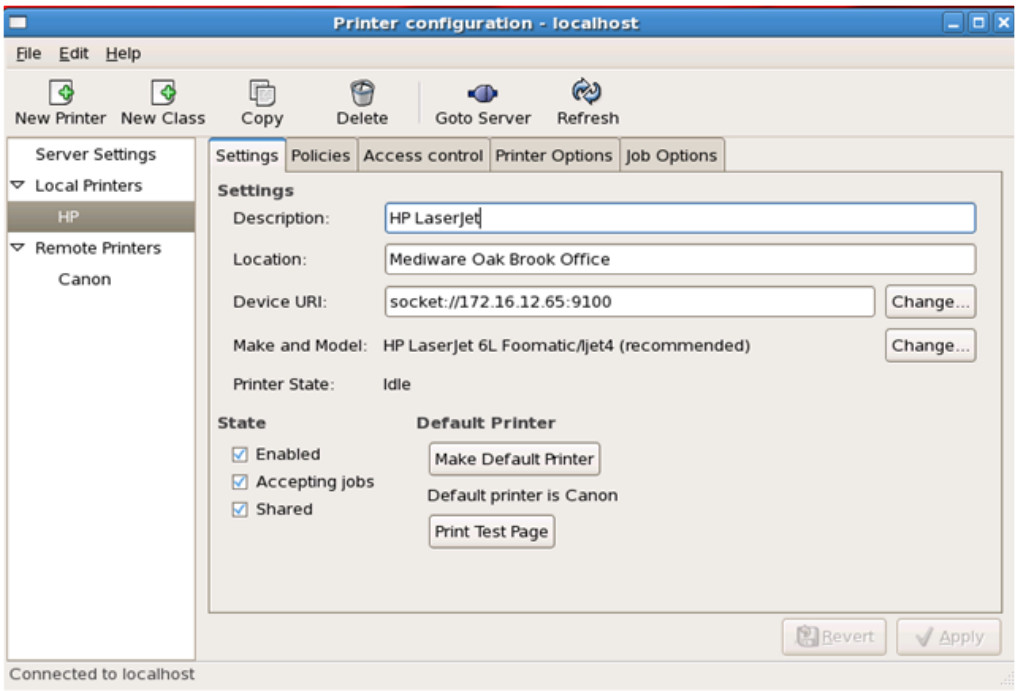
10. Click **Forward**. The "Going to create a new printer" message appears, as shown in the following example:



11. Click **Apply**. The following window opens:



12. Under Server Settings, Local Printers, select the new server printer you have just installed. The following window opens:



13. Click **Print Test Page**.

14. Verify that the test page prints correctly.

7.4 Printer Operational Checks

If users are unable to print, follow these steps to troubleshoot the problem.

1. Verify that the device is operational.
2. Cycle power to clear existing error conditions.
3. Watch for error messages on restart.
4. Print a test page from the printer's test menu to establish functionality within the unit.
5. Verify that the printer is connected to the network, user's PC, etc.
6. Print a test page using the simplest communication protocol possible, such as a Windows test page, notepad, copy, lp, etc. to establish basic printer functionality and connectivity.
7. Reboot the PC to restart the print spooler and clear error conditions. Print spooler failure will frequently result in printers appearing offline, or in defined printers disappearing from the Windows printer list until the system is rebooted.
8. Sometimes a printer definition needs to be removed and recreated to clear a problem with an individual printer.



NOTE – Reboot between removal and recreation of the printer.

7.5 Common Windows Printing Problems

Problem	Probable Cause	Recommended Solution
Print jobs are not being sent from the server print queue to the printer. A print job with a size of 0 bytes is at the top of the print job list for the printer. Other documents appear on the print job list, and users can still send print jobs to the printer. There is plenty of free space on the partition that contains the spool folder.	The most likely cause is a stalled print spooler.	Stop and restart the Spooler service.
Users can't connect a Windows computer to a printer that uses TCP/IP and LPD.	The most common problem is incorrect configuration of TCP/IP on either the Windows computer or on the printer.	Verify that the IP address, subnet mask, and default gateway parameters on both the Windows computer and the printer are correct.
A printer malfunction occurs in the middle of a print job. You want to reprint the entire print job.	Printer jam. Power outage.	1.Double-click on the printer in the Printers folder. 2.The Printers dialog box appears. Select **Document > Pause**. 3.Fix the printer malfunction. 4.Select **Document > Restart** to reprint the entire print job. **Important!** Do not select **Resume** from the Document menu because the job will print from the point where the printer malfunctioned and some pages will be lost.
You receive spooler messages indicating that your print job has been spooled with a size of 0 bytes.	The most common cause is that the printer's spool folder does not have enough space for the document.	Delete files from the partition to free space, or move the spool folder to a partition with more free space.

7.6 Clearing Windows Server Print Queues

The Windows System Administrator can kill problem print jobs by removing them from the print queue at the server. In some cases, the print spooler may need to be shut down and restarted after the job has been removed.

7.7 Checking Print Queue Space

Print failures may occur if there is insufficient space to queue print files. In this case, more space must be made available. The short-term remedy is to clear out the print queue. The long-term remedy is to locate the print queue directory so that more space is reserved for print files.

7.8 Purging Blood Centers Report Files

Blood Centers does not automatically delete user-generated reports from the reports share. The System Administrator must periodically delete old report files from the system to avoid running low on disk space and to maintain system performance and manageability. Blood Centers customers determine the retention period for reports and the frequency of file maintenance according to the needs of their site. The process can be automated if desired. The following procedure describes how to purge the report files.

1. Log onto the Blood Centers Application server as **oracle**.
2. Right-click on the desktop and select **Open Terminal**.
3. To purge old report files, enter the following command:

```
$ find /home/bcorapp/repout -name '*.pdf' -mtime +6 -exec rm -rf {} \;
```



NOTE – The command above removes report files that are more than 6 days old (in the +6 section of the command). To keep reports for a longer period, change the number. For example, to keep reports for 30 days, change the 6 to 30.

To purge old reports automatically, create a file containing the purge commands from step 3, and then add the file to a cron job and set it to run daily.

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Chapter 8 - Database Recovery Procedures

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8.1 Backup

Important! The backup and recovery scenarios that follow should only be undertaken by an Oracle professional. These guides are only to show the ways of taking backups.

8.1.1 Cold Backup

1. A COLD Backup is an offline backup, therefore the database must be down.
2. The database can be put into Archive log or No Archive Mode.

   NOTE – If it is in archive mode, recovery is possible; otherwise only restore is possible.

3. Login to the database and check the control, redolog, data and temp file locations by entering the following commands:
   ```
   $ . oraenv
   ```

   When prompted, enter the name of the database where information is to be verified (ex: prddb).

   ```
   $ sqlplus "/as sysdba"
   ```

   At the SQL prompt, enter the following commands:
   ```
   SQL> select name from v$datafile;
   SQL> select name from v$tempfile;
   SQL> select name from v$controlfile;
   SQL> select name from v$logfile;
   ```


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Example:

```
[oracle@bcqadbol95 ~]$
[oracle@bcqadbol95 ~]$ . oraenv
ORACLE_SID = [oracle] ? qasdb
The Oracle base has been set to /apps/product/19.0.0
[oracle@bcqadbol95 ~]$ sqlplus "/as sysdba"

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Jul 9 17:26:13 2025
Version 19.27.0.0.0

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Connected to:
Oracle Database 19c Standard Edition 2 Release 19.0.0.0.0 - Production
Version 19.27.0.0.0

SQL> select name from v$datafile;

NAME
--------------------------------------------------------------------------------
/oradata/QASDB/system01.dbf
/oradata/QASDB/bc_data02.dbf
/oradata/QASDB/sysaux01.dbf
/oradata/QASDB/undotbs01.dbf
/oradata/QASDB/bc_data01.dbf
/oradata/QASDB/users01.dbf
/oradata/QASDB/bc_data03.dbf
/oradata/QASDB/bc_data04.dbf
/oradata/QASDB/bc_data05.dbf
/oradata/QASDB/bc_data06.dbf
/oradata/QASDB/bc_data07.dbf

NAME
--------------------------------------------------------------------------------
/oradata/QASDB/bc_data08.dbf
/oradata/QASDB/bc_data09.dbf
/oradata/QASDB/bc_data10.dbf
/oradata/QASDB/QAS_ias_opss.dbf
/oradata/QASDB/QAS_VMS.dbf
/oradata/QASDB/QAS_iau.dbf
/oradata/QASDB/QAS_mds.dbf
/oradata/QASDB/QAS_svctbl.dbf
/oradata/QASDB/QAS_wlsservices.dbf

20 rows selected.

SQL> select name from v$tempfile;

NAME
--------------------------------------------------------------------------------
/oradata/QASDB/temp01.dbf
/oradata/QASDB/QAS_iastemp.dbf

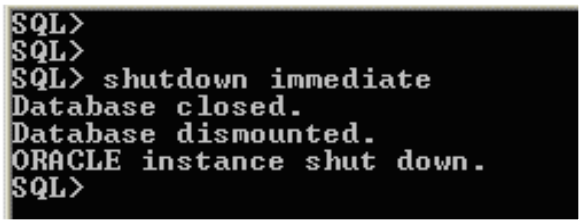
SQL> select name from v$controlfile;

NAME
--------------------------------------------------------------------------------
/oradata/QASDB/control01.ctl
/oradata/QASDB/control02.ctl

SQL> select member from v$logfile;

MEMBER
--------------------------------------------------------------------------------
/oradata/QASDB/redo03.log
/oradata/QASDB/redo02.log
/oradata/QASDB/redo01.log
```

4. Enter the following command to shutdown the database with the immediate option:
   SQL> `shutdown immediate`



5. Create a backup location and copy the physical files (datafiles, redologs, control, and temp files) to the backup location.
   a. Create the backup directory with date format – $ `mkdir /DB_BKPS/<Date>`
   b. In our example, all physical files are under the /oradata/qasdb directory. Copy the /oradata/qasdb directory to the backup location – $ `cp –rf /oradata/qasdb /DB_BKPS/<Date >`
6. Copy the archive logs to the backup location.
7. Copy the backup from the backup location to backup media, and purge it according to file system space.

8.1.2 Hot Backup

1. Hot backup is an online backup, therefore the database is not required to be shutdown.

   NOTE – While doing the Hot Backup, users can perform read/write operations on the database since Hot Backup is transparent to users.

2. The database should be in Archive log Mode.
3. The database needs to be in backup mode.

   NOTE – You can take only data files, traced controlfile, and archive files backup. There is no redo log file backup in a hot backup.

4. Login to the database and select the datafiles location by entering the following commands:
   $ `sqlplus "/as sysdba"`

   At the SQL prompt, enter the following command:
   SQL> `Select name from v$datafile;`
5. Alter database in backup mode by entering the following command:
   SQL> `alter database begin backup;`
   SQL> `exit`
6. Copy the data files into the backup location using the following command:
   $ `cp –rf /oradata/qasdb/*.dbf /DB_BKPS/<date>/qasdb/`

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NOTE – Replace qasdb in the command above with the name of your database.

7. Alter the database to end backup mode by entering these commands:
   $ `sqlplus "/as sysdba"`

   At the SQL prompt, enter the following command:
   SQL> `alter database end backup;`
8. Switch the log file manually, so that the updated information will be in the archive log location.
   SQL> `alter system switch logfile;`
   SQL> `exit`
9. Backup all archive logs generated to the backup location and purge the archive location according to space by entering the following command:
   $ `cp -rf /opt/oracle/admin/qasdb/arch/ /DB_BKPS/<Date>/qasdb/arch`

   NOTE – Replace qasdb in the command above with the name of your database.
10. Backup the control file to the backup location in asci format by entering the following command:
    $ `sqlplus "/as sysdba"`

    At the SQL prompt, enter the following command:
    SQL> `alter database backup controlfile to trace as '/DB_BKPS/<Date>/qasdb/qasdb_ctl.sql';`

    NOTE – Replace qasdb in the command above with the name of your database.
11. Take the backup from the backup location to backup media, and purge it according to file system space.

8.2 ORACLE Instance Recovery

When a Blood Centers Oracle database instance fails, the customer should contact Blood Centers Support for assistance according to the terms of the support contract.

Important! Execute the database recovery procedures described in this section only under the expert guidance of Blood Centers Support personnel.

The ORACLE instance provides the software mechanism for accessing and controlling the database. It consists of a shared memory area and a set of processes that start as soon as an instance starts.

An Oracle instance that cannot continue to function, for any reason, is in "instance failure". Oracle provides a mechanism for automatic recovery. Only a qualified DBA should execute this recovery procedure under the direction of Blood Centers Support personnel. The DBA proceeds as follows:

- Use the sqlplus command 'SHUTDOWN IMMEDIATE' to shut down the instance.
- Use the sqlplus command 'STARTUP' to restart the instance. Oracle executes recovery automatically on restart. NO OPERATOR INTERVENTION IS REQUIRED.

Important! Recovery must be performed only under the direction of a qualified DBA in accordance with procedures outlined in the Oracle Administrator Manual.

8.3 Media Recovery Procedures

This section describes database recovery for a variety of conditions related to loss of files and recovery from backup media.

8.3.1 Recover from loss of single control file

1. Copy one of the other control files.
2. Restart the database.

8.3.2 Recover from loss of all control files

1. Mount the database using sqlplus.
   SQL> `startup mount;`
2. Restore the backup control file from tape.
3. Recover the database.
   SQL> `recover database using backup controlfile;`
4. Oracle will recommend the restore of archived log files.
5. Oracle prompts for an archived log file that apparently does not exist physically. This log file is actually the current redo log, which is to be applied to complete the recovery.
6. Open the database.
   SQL> `alter database open resetlogs;`
7. The database starts normally.

NOTE – Database archived redo log files are reset automatically at startup. To protect your system, you must take a complete online or offline database backup immediately after this type of recovery.

Important! If the complete database backup is not performed, a new start sequence resets and clears the redo logs to date. Recover from loss of rollback table space data file.

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8.3.3 Recover from loss of system table space data file

1. Restore the data file from the latest backup.
   $ `cd /usr01`
   $ `cpio -ivd sys*.dbf </dev/st0 &`
2. When the copy is complete, mount the database using sqlplus but do not open it.
   SQL> `startup mount;`
3. Recover the data file.
   SQL> `recover datafile <filename>;`
   Oracle displays "Media recovery complete" after applying the log files.
4. Open the database.
   SQL> `alter database open;`
   The database starts normally.

8.3.4 Recover from loss of table space data file

1. Shut down the database using sqlplus.
   SQL> `shutdown abort;`

   NOTE – Normal shutdown will not work.

2. Mount the database, but do not open it.
   SQL> `startup mount;`
3. Determine which files need database recovery.
   SQL> `select * from v$recover_file;`
   The command shows a list of the files needing recovery. Verify that the name of the deleted file is displayed.
4. Restore the data file from the latest tape backup.
   $ `cd /`
   $ `cpio -ivd <filename> </dev/st0 &`
5. Open the database.
   SQL> `alter database open;`

   NOTE – This type of recovery enables the database to be opened for use; only files needing to be recovered are unavailable. If the file to be recovered is heavily accessed by users, do not open the database for use. "Recover database" is a better option, in which the database is mounted but not opened.

6. Recover the data file.
   SQL> `recover datafile <filename>;`

   Apply redo logs or user “auto”. The archived redo logs are assumed to be found in the archived redo log location. If not, then perform media recovery of archived redo logs.
7. Take the data file online.
   SQL> `alter database datafile <filename> online;`
8. Check that the database is OK.
   SQL> `shutdown;`
   SQL> `startup;`
   The database starts normally.

8.3.5 Recover from loss of temporary table space data file

1. Open the database using sqlplus.
   SQL> `startup mount;`
   SQL> `alter database <filename> offline;`
   SQL> `alter database open ;`
2. Take the temporary table space offline.
   SQL> `alter tablespace temp offline;`
3. Drop and recreate the tablespace. The database can function without the temporary tablespace.
   SQL> `drop tablespace TEMP including contents;`
   SQL> `create tablespace temp datafile <filename>;`
   The database should function normally.

8.3.6 Recover from loss of all data files for a single table space

1. Shut down the database using sqlplus.
   SQL> `shutdown abort;`

> **NOTE –** Normal shutdown will not work.

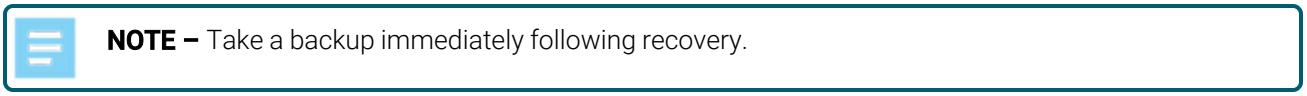
2. Mount the database but do not open it.
   SQL> `startup mount;`
3. Set specific data files offline:
   SQL> `alter database datafile <filename> offline;`
4. Open the database.
   SQL> `alter database open;`
5. Make tablespace to be recovered offline.
   SQL> `alter tablespace <Table Space Name> offline;`

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6. Restore the data files from the latest tape backup.
   $ `cd /`
   $ `cpio -ivd </dev/st0 &`
7. Recover the tablespace.
   SQL> `recover tablespace <Table Space Name>;`
8. Bring the tablespace online.
   SQL> `alter tablespace <Table Space Name> online;`
9. Check that the database is OK.
   SQL> `shutdown;`
   SQL> `startup;`
   The database starts normally.

8.3.7 Recover from loss of redo log file

1. If the error is due to a hardware problem on one member of a mirrored redo file, fix the hardware problem and the error will be resolved automatically.
2. If the hardware problem is permanent, drop the log file member and add another member.
3. If all the members of an inactive redo group are damaged:
   - If the problem is noticed before the database shutdown, use sqlplus to clear the log file.
     SQL> `alter database clear logfile ;`
   - If the database is already shutdown due to the problem, use
     SQL> `startup mount;`
   - If the logfile was archived,
     SQL> `alter database clear logfile;`
   - Otherwise
     SQL> `alter database clear unarchived logfile;`



4. If the clear logfile command fails, then add another member to the log file group and drop the bad member.
5. If all members of the active log file group are lost, a complete media recovery must be performed.
   a. Perform recovery using the log files up to the time the damaged log file was opened.
   b. Create new log files and drop the bad files.

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